New answer metrics for the simulated environment

Information Collection Strategies Experiment · environment benchmark · 13 October 2026

> **What was asked.** The two existing answer metrics (masked correctness and factual consistency) barely separate the retrieval methods. The task was to add metrics that separate them more, test each on one run with one judge, revise once if needed, then score all ten stored runs with the whole judge panel, with the Jev decision model as an added judge.

- **Faithfulness separates every pair of retrieval methods.** Its spread between methods is 8.1 times its run-to-run noise; for the two old metrics the ratio is 0.2 to 0.7. The documents support 59% of Perplexity's answers, 35% with retrieval changes 1–4, and 21–25% for the other three methods.
- **The zlog z-score does not separate the methods** (ratio 0.5–0.6, like masked correctness). The methods' answers are equally close to the true value; Gemini without any search is as close as the search methods.
- **Case alignment separates them slightly** (ratio 1.0–1.4): answers from Gemini's own knowledge and from Perplexity are contradicted by the case a little more often (about 3% against 1.5%). The differences are one to two points.
- **The old metrics hardly moved under the new panel** (Claude Haiku 5.5 and GLM 5.3 Flash in place of Gemini and Qwen): same category 83–86%, consistent about 97%, as before.
- **Adding Jev as a fourth judge changes little:** most scores stay within one point. Exact masked correctness falls by one to two points, and native Google search's faithfulness by three (23.5% to 20.6%, one run).
- **One part is incomplete.** Faithfulness and case alignment by the three chat judges cover nine runs, not ten: the API key reached its total spending limit of $250 during run 10. Everything else covers all ten runs.

Final scores: three judges

Panel: Claude Haiku 5.5, GLM 5.3 Flash, GPT-6 Luna; the label is the majority vote, the zlog score the median of the judges. Mean ± SD over runs. Green headers are the new metrics. ↓ lower is better. Ten runs, except faithfulness and case alignment (nine runs) and the two native search configurations (one run).

Configuration	Masked correctness		Factual consistency	Zlog z-score		Faithfulness	Case alignment	
	same category (incl. exact)	exact	consistent	mean |z| ↓	satisfactory	supported	aligned	contradicted ↓
Gemini 3 Flash knowledge, no search	84.4 ± 1.5%	45.6 ± 1.4%	97.3 ± 1.5%	1.28 ± 0.04	74.6 ± 2.0%	–	18.6 ± 0.9% (9)	3.3 ± 0.5% (9)
Europe PMC + LitSense, without changes 1–4	86.2 ± 1.5%	45.4 ± 2.5%	96.8 ± 1.2%	1.26 ± 0.05	73.0 ± 1.8%	22.7 ± 1.3% (9)	20.0 ± 1.1% (9)	1.7 ± 0.6% (9)
Europe PMC + LitSense, with changes 1–4	84.9 ± 1.3%	41.6 ± 2.1%	96.8 ± 1.1%	1.29 ± 0.05	73.1 ± 2.8%	35.1 ± 2.5% (9)	20.1 ± 0.6% (9)	1.5 ± 0.5% (9)
OpenRouter search: Exa instant	85.0 ± 1.5%	45.9 ± 3.0%	97.2 ± 0.9%	1.25 ± 0.06	75.4 ± 2.7%	21.0 ± 1.3% (9)	20.2 ± 0.8% (9)	1.4 ± 0.6% (9)
OpenRouter search: Parallel basic	84.0 ± 2.2%	43.4 ± 2.6%	96.7 ± 1.0%	1.29 ± 0.06	71.9 ± 2.8%	25.4 ± 1.8% (9)	20.3 ± 1.0% (9)	1.3 ± 0.7% (9)
OpenRouter search: Perplexity	83.2 ± 2.3%	43.9 ± 2.7%	97.1 ± 0.6%	1.32 ± 0.04	72.1 ± 2.1%	59.2 ± 2.4% (9)	18.2 ± 0.8% (9)	2.7 ± 0.7% (9)
Native Google search (one run)	82.3%	44.7%	96.7%	1.25	71.4%	23.5%	19.1%	1.8%
Native OpenAI search (one run)	84.4%	44.0%	95.6%	1.28	72.9%	38.3%	19.5%	1.1%
Case study only, no retrieval (set C)	99.2 ± 0.4%	98.8 ± 0.4%	99.7 ± 0.3%	0.04 ± 0.02	98.7 ± 0.6%	–	97.7 ± 0.8% (9)	0.3 ± 0.3% (9)

Final scores: the same table with the Jev decision model added as a fourth judge

Panel: the three judges and Jev 1.13; with a 2–2 split the main judge (Claude Haiku 5.5) decides. Jev does not take part in the zlog z-score, so those two columns are the same as above. Runs as above.

Configuration	Masked correctness		Factual consistency	Zlog z-score		Faithfulness	Case alignment	
	same category (incl. exact)	exact	consistent	mean |z| ↓	satisfactory	supported	aligned	contradicted ↓
Gemini 3 Flash knowledge, no search	83.8 ± 1.6%	43.7 ± 2.1%	97.2 ± 1.5%	1.28 ± 0.04	74.6 ± 2.0%	–	19.1 ± 1.1% (9)	3.1 ± 0.7% (9)
Europe PMC + LitSense, without changes 1–4	86.0 ± 1.4%	44.2 ± 2.8%	96.8 ± 1.2%	1.26 ± 0.05	73.0 ± 1.8%	21.9 ± 1.3% (9)	20.8 ± 1.2% (9)	1.4 ± 0.7% (9)
Europe PMC + LitSense, with changes 1–4	84.7 ± 1.5%	40.3 ± 2.5%	96.8 ± 1.1%	1.29 ± 0.05	73.1 ± 2.8%	35.0 ± 1.9% (9)	21.0 ± 0.6% (9)	1.1 ± 0.3% (9)
OpenRouter search: Exa instant	84.1 ± 1.8%	44.8 ± 3.2%	97.1 ± 0.9%	1.25 ± 0.06	75.4 ± 2.7%	20.3 ± 1.2% (9)	21.3 ± 1.0% (9)	1.1 ± 0.7% (9)
OpenRouter search: Parallel basic	84.5 ± 2.3%	41.9 ± 2.6%	96.7 ± 1.0%	1.29 ± 0.06	71.9 ± 2.8%	24.1 ± 1.7% (9)	21.0 ± 0.7% (9)	1.2 ± 0.7% (9)
OpenRouter search: Perplexity	82.9 ± 2.4%	42.8 ± 2.9%	97.1 ± 0.7%	1.32 ± 0.04	72.1 ± 2.1%	59.5 ± 2.3% (9)	19.2 ± 0.7% (9)	2.4 ± 0.6% (9)
Native Google search (one run)	82.3%	43.3%	96.7%	1.25	71.4%	20.6%	20.2%	1.8%
Native OpenAI search (one run)	83.7%	41.8%	95.6%	1.28	72.9%	37.8%	20.2%	1.1%
Case study only, no retrieval (set C)	99.2 ± 0.4%	98.8 ± 0.5%	99.4 ± 0.4%	0.04 ± 0.02	98.7 ± 0.6%	–	97.1 ± 0.9% (9)	0.7 ± 0.2% (9)

What the scores show

- **Retrieval changes what an answer rests on, not how right it is.** Faithfulness ranges from 21% to 59%, while every closeness measure is flat: same category 83–86%, mean |z| 1.25–1.32.
- **Changes 1–4 do what they were built for.** They raise supported answers from 22.7% to 35.1% for the same sources, although masked correctness did not change.
- **Perplexity is the most grounded and Exa instant the least** (59.2% against 21.0%). Native OpenAI search reaches 38.3% and native Google search 23.5%, each on one run.
- **The judges confirm less support than the environment claims.** It labels 61% of Perplexity's answers, 47% with changes 1–4, and about 30% of the others as taken from the documents; the judges confirm 59%, 35%, and 21–25%.
- **Case alignment:** about one answer in five is aligned with something the case says and 1–3% are contradicted, for every retrieval method. The rest are neutral: the case gives no expectation for the variable.
- **Set C behaves as it should.** When the case states the answer, mean |z| is 0.04 and 97.7% of answers are aligned.
- **Empty cells:** faithfulness has no value for Gemini knowledge and for case study only, which retrieve no documents.

Why the old metrics separate the methods poorly

- **Coarse labels.** Masked correctness has four labels and factual consistency two. Turning a continuous outcome into categories loses at least a third of its statistical information [1, 2].
- **A ceiling.** About 97% of set B answers are consistent, for every method.
- **They never look at the retrieved documents.** When the documents are weak, the synthesizer answers from its own knowledge and still lands in the right category.

The three new metrics

	Zlog z-score	Faithfulness	Case alignment
Question	How far is the answer's value from the hidden true value?	Do the retrieved documents support the answer's value?	Does the rest of the case bear on the answer, and which way?
Sets	A (140 of 141 questions have a numeric true value) and C	A and B	B and C
Based on	The zlog transformation of laboratory results [3] and the proficiency-testing z-score of ISO 13528 [4]	Faithfulness in the evaluation of retrieval-augmented generation [5, 6]	The three labels of natural language inference, the standard check of a text against its source [7, 8]
Judge	Reads numbers only: once per question the reference limits and the true value, once per answer the answer's value in the true value's unit	Yes or no, with the supporting passage quoted word for word	Reads the level the case leads one to expect and the level the answer states, with the case passage that grounds the expectation
Code	Computes z; the answer's score is the median of the judges' scores	Looks for the quote in the documents; not found means not supported (1–4% of the "yes" verdicts)	Looks for the quote in the case; compares the two levels
Reported	Mean |z| capped at 3, and the share satisfactory (|z| ≤ 2). Unanswered questions score 3	Share of answers supported, of all questions	Share aligned and share contradicted, of all questions

Zlog z-score: the equations

```
zlog(x) = (ln x − (ln LL + ln UL)/2) · 3.92 / (ln UL − ln LL)     [3], LL and UL: lower and upper reference limits
```

```
z = zlog(answer) − zlog(truth) = (ln answer − ln truth) / σ,   σ = (ln UL − ln LL) / 3.92     [4], the true value as the assigned value
```

```
revised (kept): σ is multiplied by max(1, |zlog(truth)| / 3.92)
```

The answer's error is measured in units of the healthy population's spread. ISO 13528 rates |z| ≤ 2 satisfactory and |z| ≥ 3 unsatisfactory. Rules beyond the two methods, all this project's choices: a one-sided reference range takes a lower limit of 15% of the upper; a variable without a reference interval takes σ = 0.15; values at or below zero use the same z without logarithms; for blood pressure the systolic value is scored; a true value given as a range or bound scores 0 for an answer inside it.

Development: the test on one run with one judge

Run 2, Claude Haiku 5.5 as the only judge, the six configurations that were run ten times (five for faithfulness). **Spread** is the standard deviation of the configurations' means; **noise** is their mean standard error from a bootstrap that resamples diagnoses; **pairs separated** counts the pairs of configurations whose paired 95% interval excludes zero. A metric is more sensitive when signal ÷ noise is higher and more pairs are separated. Percent metrics are in points.

Metric	Version	Gemini knowledge	EPMC + LitSense, without 1–4	EPMC + LitSense, with 1–4	Exa instant	Parallel basic	Perplexity	Spread	Noise	Signal ÷ noise	Pairs separated
Masked correctness: same category (incl. exact), %	old	83.0	87.2	85.1	84.4	87.9	81.6	2.44	3.16	**0.77**	2 of 15
Masked correctness: exact, %	old	44.0	44.0	36.2	46.8	41.8	44.0	3.61	4.37	**0.83**	1 of 15
Factual consistency: consistent, %	old	96.7	97.1	95.6	97.1	97.4	96.3	0.66	1.09	**0.61**	0 of 15
Zlog z-score, first version: mean |z|	first	1.4	1.4	1.4	1.4	1.4	1.4	0.02	0.10	**0.23**	0 of 15
Zlog z-score: mean |z| (capped at 3)	revised, kept	1.2	1.2	1.2	1.2	1.2	1.2	0.02	0.09	**0.25**	0 of 15
Zlog z-score, first version: satisfactory, %	first	71.4	67.9	67.1	67.9	67.9	67.9	1.54	4.03	**0.38**	0 of 15
Zlog z-score: satisfactory (|z| ≤ 2), %	revised, kept	78.6	77.9	74.3	75.7	74.3	77.1	1.83	3.78	**0.48**	0 of 15
Faithfulness: supported, %	kept as designed	–	16.9	29.5	15.5	22.8	56.7	16.80	2.09	**8.03**	9 of 10
Case alignment, first version: aligned, %	first	3.3	5.1	3.7	4.8	4.4	4.0	0.69	1.20	**0.57**	0 of 15
Case alignment: aligned, %	revised, kept	15.8	16.5	19.1	18.4	17.3	18.8	1.32	2.48	**0.53**	0 of 15
Case alignment, first version: contradicted, %	first	0.0	0.0	0.0	0.0	0.0	0.0	0.00	0.00	**0.00**	0 of 15
Case alignment: contradicted, %	revised, kept	3.7	2.6	2.2	1.8	2.9	2.6	0.63	1.06	**0.60**	0 of 15

- **Faithfulness** was kept as designed: about ten times the old metrics' ratio.
- **Zlog z-score, one revision.** 22% of scores sat at the cap, many for very abnormal true values answered with similarly abnormal ones (haemoglobin 4 g/dL answered as 6.8). The revised scale widens the tolerance with the true value's own distance from normal. It was kept (0.25 against 0.23), a difference well inside the noise.
- **Case alignment, one revision.** The first version asked for one verdict and counted only explicit statements: 96% of answers came back neutral and none contradicted. The revised version lets the clinical picture set an expectation and compares two levels. It was kept (0.60 against 0.57), again inside the noise, but it uses the scale: 16–19% aligned, 2–4% contradicted.
- Neither revision made its metric more sensitive than the old ones on this run. The nine-run figures below show the same for the zlog z-score and a modest gain for case alignment.

Sensitivity over all judged runs

Spread is the standard deviation of the configurations' means over runs, **run SD** the pooled run-to-run standard deviation, and **pairs separated** the pairs whose means differ by more than 2.3 standard errors (Welch's t). Percent metrics are in points.

Metric	Three judges					With Jev added				
	Runs	Spread	Run SD	Signal ÷ noise	Pairs separated	Runs	Spread	Run SD	Signal ÷ noise	Pairs separated
Masked correctness: same category (incl. exact), %	10	1.02	1.77	**0.58**	3 of 15	10	1.01	1.86	**0.54**	3 of 15
Masked correctness: exact, %	10	1.67	2.44	**0.68**	4 of 15	10	1.66	2.74	**0.61**	3 of 15
Factual consistency: consistent, %	10	0.24	1.08	**0.22**	0 of 15	10	0.24	1.09	**0.22**	0 of 15
Zlog z-score: mean |z| (capped at 3)	10	0.03	0.05	**0.52**	2 of 15	10	0.03	0.05	**0.52**	2 of 15
Zlog z-score: satisfactory (|z| ≤ 2), %	10	1.39	2.42	**0.58**	5 of 15	10	1.39	2.42	**0.58**	5 of 15
Faithfulness: supported, %	9	15.80	1.94	**8.13**	10 of 10	9	16.32	1.71	**9.54**	10 of 10
Case alignment: aligned, %	9	0.93	0.88	**1.05**	8 of 15	9	0.98	0.92	**1.07**	8 of 15
Case alignment: contradicted, %	9	0.83	0.59	**1.39**	9 of 15	9	0.84	0.65	**1.29**	8 of 15

The judges

- **Panel:** Claude Haiku 5.5 (main judge, breaks ties), GLM 5.3 Flash, and GPT-6 Luna. Haiku replaced Gemini 3.5 Flash Lite and GLM replaced Qwen 3.8 Flash, in the old scores as well: both re-judged every answer of the ten runs. GLM was chosen from four candidates on 40 stored answers.
- **Jev 1.13** is a decision model: it returns a choice with probabilities and no text. It judged both old metrics, faithfulness, and case alignment for all ten runs. It cannot read out a number, so the zlog z-score has three judges in both tables, and its verdicts cannot be quote-checked.
- **Agreement on faithfulness** (18,904 answers): the three judges are unanimous on 75.5%; any two agree on 81–85%, and Jev agrees with each of them as often (81–85%).
- **Agreement on case alignment** (14,823 answers): the three judges are unanimous on 79.3% and agree pairwise on 84–88%. Jev agrees with them on only 60–67%: it calls 53% of answers aligned, the others 17–27%. With three other votes it rarely decides the label.
- **Batch API:** not used. Gemini's batch endpoint is refused under the account's data-retention setting, and two GPT batches had completed nothing after 25 minutes.
- **Volume and cost:** about 289,000 judgments; $56.60 spent on the day, including tests and calibration.

Limits

- **Nine runs, not ten, for two metrics.** The API key's total limit of $250 was reached during run 10, so the three chat judges' faithfulness and case alignment for that run are unfinished and left out. Jev's verdicts for run 10 exist but are not shown alone.
- **Faithfulness rewards grounding, not truth.** An answer can be faithful to a document that is wrong for this patient, and a method that returns more text has more chances of containing an agreeing number.
- **The planned sanity check for faithfulness was not run** (another question's documents should give "not supported"); the key limit was reached first.
- **The versions were chosen on one run and one judge**, on differences smaller than the noise.
- **The reference limits come from the judges**, as the median of three; 103 of the 140 set A questions have a two-sided interval, 15 a one-sided one, 22 none.
- **The revised zlog scale and the special-case rules are this project's own**, not from the cited methods.
- **All judges are language models.** No human labels were collected for the new metrics.
- **Native Google and OpenAI search** have one run each, so no standard deviation.

References

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Data and code: environment/results/answer_metrics/ (scores, test, logs), environment/bench/answer_metrics.py, environment/bench/zlog.py, environment/results/scripts/make_answer_metrics.py. Method details: environment/bench/README.md.